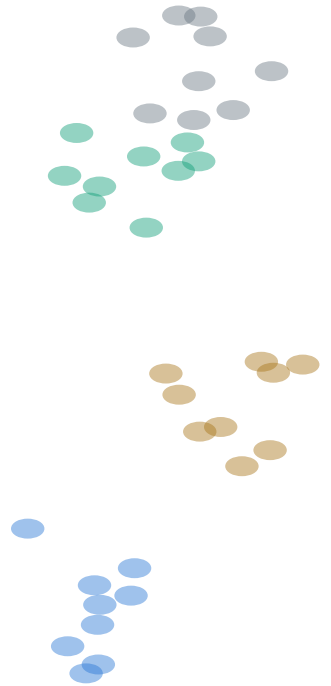
Physics-Defined Tokenization for AI-Ready HEP Datasets

Combined findings from six experimental campaigns, and a ranked assessment of tokenization schemes for the released token.



heptokens working group

Data: CMS Aspen Open Jets RunG (1.96 M jets) · 620 k-jet 5-class benchmark

August 2026

OUTLINE

Outline

1 Introduction — the AI-ready-dataset mission and the tokenizer question

2 Methodology — data, fairness protocol, and what we measure (and why)

3 Combined findings — six results, evidence pooled across all campaigns

4 Rankings — what works, what to test next, and what is closed

5 Conclusions

Motivation: the token dataset is the product

- Experiments reconstruct particle-flow candidates that external groups cannot access; those groups want to train foundation models on the events. We publish a per-constituent token as the shareable artifact — versioned by dimension, preprocessing hash and checkpoint.
- Release requirements: information sufficiency for unknown future tasks; detector abstraction; an auditable statement of what the token forgets; compression at the 178 M-jet scale.
- Starting point: a residual VQ-VAE emitting integer indices — with codebook collapse, an implicit metric set by the reconstruction objective, and lossiness by construction. Six campaigns (2026-07 → 2026-08) tested the alternatives; this talk presents the combined outcome.

RESEARCH QUESTION

Which objective and which bottleneck should define the released per-particle token —
and can any single token be simultaneously compact, information-faithful, and criterion-organized?

Data, the library objective, and the fairness protocol

THE PHYSICS CRITERION (defines 'same-physics particle')

positives: same PID ∧ same charge ∧ ΔpT/pT ≤ 0.10 ∧ |Δη| ≤ 0.1 ∧ |Δφ| ≤ 0.1 (graded kernel, hard PID/charge gates)

- Data: Aspen Open Jets RunG — 1.96 M real CMS jets, constituents (pT, η, φ, PID, charge); all criterion results. jets5c (620 k labelled jets) for downstream consumer probes.
- Matched budgets everywhere: identical encoders, inputs, splits, schedules; only the objective/bottleneck under test differs. Context-free per-particle MLP unless stated.
- Consumer view: downstream probes see only the exported token files, never the constituents.
- Untrained nulls are mandatory (PID-one-hot ⊕ kinematics; criterion-aligned feature grid): learning must beat them to claim rent. 3 seeds wherever a number becomes a claim.

single A100 MIG slice · torch 2.5.1 · 15-20 epochs per arm · full configs in benchmarks/

Model architectures under test, and the library baseline

HEPTOKENS LIBRARY BASELINE — the pre-study production tokenizer (LitVqVae)

`transformer encoder (4 layers, d_model 128)  →  residual VQ: 3 × 512 codes @ dim 16 (STE + EMA)  →  transformer decoder`

objective: full-particle reconstruction · release = integer code indices, 27 bits/particle

kept as the reference arm in every campaign; from C2b also in matched form (vqvae_particle: the library's own encoder, only the objective differs).

ENCODERS	context-aware transformer (4 L, d_model 128 — C1; the C5 'attention' arm) vs context-free per-particle MLP ParticleEncoder (128-128-64; PID/charge embeddings) — the method of record from C2 on. Capacity study: width 64 ≈ 256 (the encoder is overbuilt); every encoder is 10^4–10^5 parameters by design.
BOTTLENECKS	none — deterministic continuous token, dim 4–16 (LitContrastiveTokenizer) · Gaussian VAE, token = posterior mean μ, β/free-bits dials (LitVae) · residual VQ 3×512 / 3×1024 / 1×8192, STE+EMA (LitVqVae) · FSQ fixed grid, 10–24 b, zero parameters · untrained: criterion feature grid, post-hoc residual k-means.
OBJECTIVES	full-particle reconstruction (L1 kinematics + CE PID + CE charge) · graded physics-kernel contrastive (method of record) · hard-window SupCon · smearing-VICReg two-view (v1) · hybrids: kernel + w·recon; kernel → z_q / z_e on VQ (LitContrastiveVqVae).
CONSUMER PROBE	2-layer transformer classifier that reads only the exported token files — never the constituents.

method of record = ParticleEncoder MLP + graded kernel, no bottleneck · all Lightning modules in src/heptokens/models/

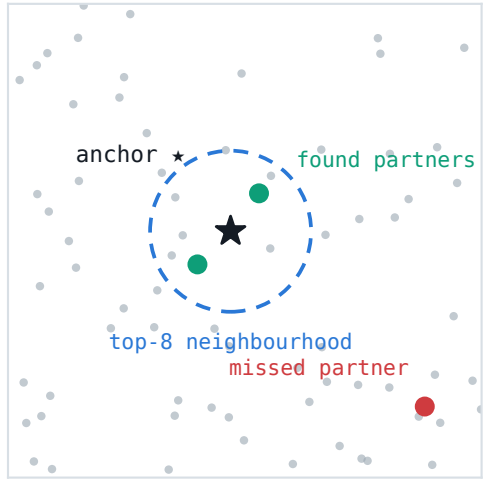
How a token is scored — a worked example

1 **Pick a particle (the anchor):**
a π^+ with pT = 10 GeV at (η, φ) = (0.5, 1.2).

2 **List its true partners, using physics only:**
every π^+ (charge +1) with pT between 9.1–11 GeV, within ±0.1 in η and φ.
In a 30,000-particle pool, typically only 1–3 such particles exist.

3 **Now use the tokens:**
rank all 30,000 particles by how close their token is to the anchor's token (a token is just a short list of numbers; 'close' = nearby point).

4 **Score:**
of the partners that exist, how many appear in the top 8 of that ranking?



token space (schematic)

PRECISION@8 — THE WRONG SCORE HERE

“how many of the 8 slots hold partners?” — with only 1 partner in existence, even a PERFECT token scores 1/8 = 0.125: the number measures how rare partners are, not how good the token is — every method looks identical.

RECALL@8 — THE SCORE WE USE

“of the partners that exist, how many were found?” — a perfect token scores 1.0 at any rarity, and 1 − recall = fraction missed: library 0.9992 vs untrained reference 0.9833 → 0.08 % vs 1.7 % of partners missed — a 20× reduction.

averaged over anchors with ≥1 partner · denominator capped at min(8, #partners): 20 partners can't all fit in 8 slots

What we measure, and why

PRIMARY: availability-aware criterion recall@8 = mean over anchors of |top-8 ∩ partners| / min(8, #partners)

criterion density is ~$3 \cdot 10^{-5}$, so precision@k is bounded by partner availability and flattens all methods — recall against the availability ceiling is 1.0 for a perfect embedding at any density. Native to the library objective; always flagged.

SUPPORTING METRIC	WHAT IT MEASURES	FAILURE IT GUARDS AGAINST
kNN PID purity	species organization (neutral)	misleads alone: 0.996 purity with 0.383 recall
within-PID pT R^2	kinematic information retention	catches PID-shortcut tokens; releases need observables
linear PID/charge probes	consumer linear decodability	kernel structure is metric, not linear (probe ≈ 0.37)
effective rank	dimensional collapse	VICReg/VQ collapse modes (quantizer-0 <10 % codes)
bits / particle	release rate	$n_q \cdot \log_2 K$ · fp16 dim·16 · KL-rate is diagnostic only (F6)
downstream probe (jets5c)	consumer value of the export	tokens are the product; probes never see constituents

No metric is trusted alone: every result table reports the suite, with the objective-native column flagged.

F1 — state the metric in physics; keep the encoder context-free

TOKEN OBJECTIVE (AOJ)	RECALL@8	KNN PURITY
library: explicit graded criterion, context-free MLP	**0.9992**	0.910
explicit criterion, hard windows	0.9935	0.664
implicit metric: smearing-VICReg + jet-context transformer	**0.052**	0.284

- The augmentation list of a generic SSL recipe silently defines the metric: smearing pT traded away pT sensitivity and cost ≈1.5 AUC on labelled downstream probes.
- Graded kernels beat hard windows on every axis — hard windows kink the geometry at the window edge.
- Jet context breaks the library property (the same particle must map to the same token in any event) and destroys low-rate tokens (recall 0.383 → 0.051); its only gain is at high rate (+0.022, 1 seed).

evidence: campaigns 1, 2, 5 · benchmarks/README.md

F2 — geometry and information are separable currencies; the corner is unreached

INTERFACE	BITS	RECALL@8	PT R^2	PID PROBE	PURITY
library dim 6 (kernel)	96	**0.9996**	0.743	0.49	0.980
contrastive VAE β=.03	96 [a]	0.9984±.0015	0.758	0.59	0.989
hybrid kernel+recon w=3	256	0.9955	**0.908**	–	0.935
recon-only VAE β=0.1	96 [a]	0.976±.004	0.920±.011	**1.00**	0.995
plain VQ 3×1024	30	0.945	**0.970**	**1.00**	0.996
null: PID-1hot ⊕ kin (untrained)	192	0.9833	0.949	1.00	0.991

- No trained token reaches the corner (recall ≥ 0.999 ∧ R^2 ≥ 0.9): the best single artifact, hybrid w=3, misses recall by 0.004; the kernel arms own geometry, the reconstructive arms own information.
- Reconstruction never installs the criterion beyond the untrained null; the physics kernel is the only objective that has beaten it (0.9984 vs 0.9833).

evidence: campaigns 2b, 3, 5, 6a · [a] fp16 release size; measured KL-rate 11-13 bits (see F6)

F3 — quantizing a criterion-shaped embedding fails: any quantizer, any rate

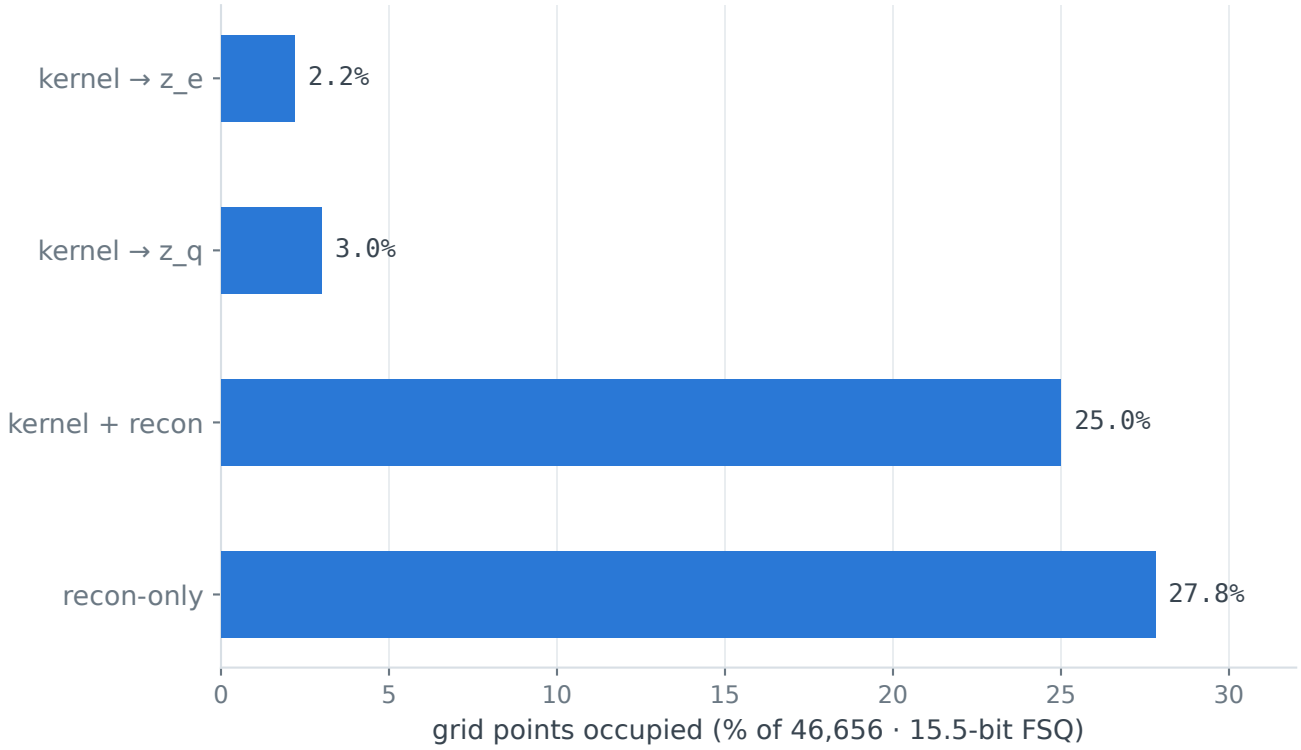


The physics kernel degrades every quantized bottleneck it touches: learned RVQ codebooks (0.945 → 0.827 at 30 b; both z_q and z_e placements) and codebook-free FSQ grids at 10, 15.5, 18 and 24 bits.

Removing the suspected culprits — codebook, EMA, commitment loss — made it worse, not better. The same kernel on a smooth continuous latent lifts recall 0.383 → 0.916.

evidence: campaigns 4, 5, 6c · kernel→z_q ablation 0.092 · kernel+recon w=1: 0.571

F3 (mechanism) — kernel-induced code degeneracy

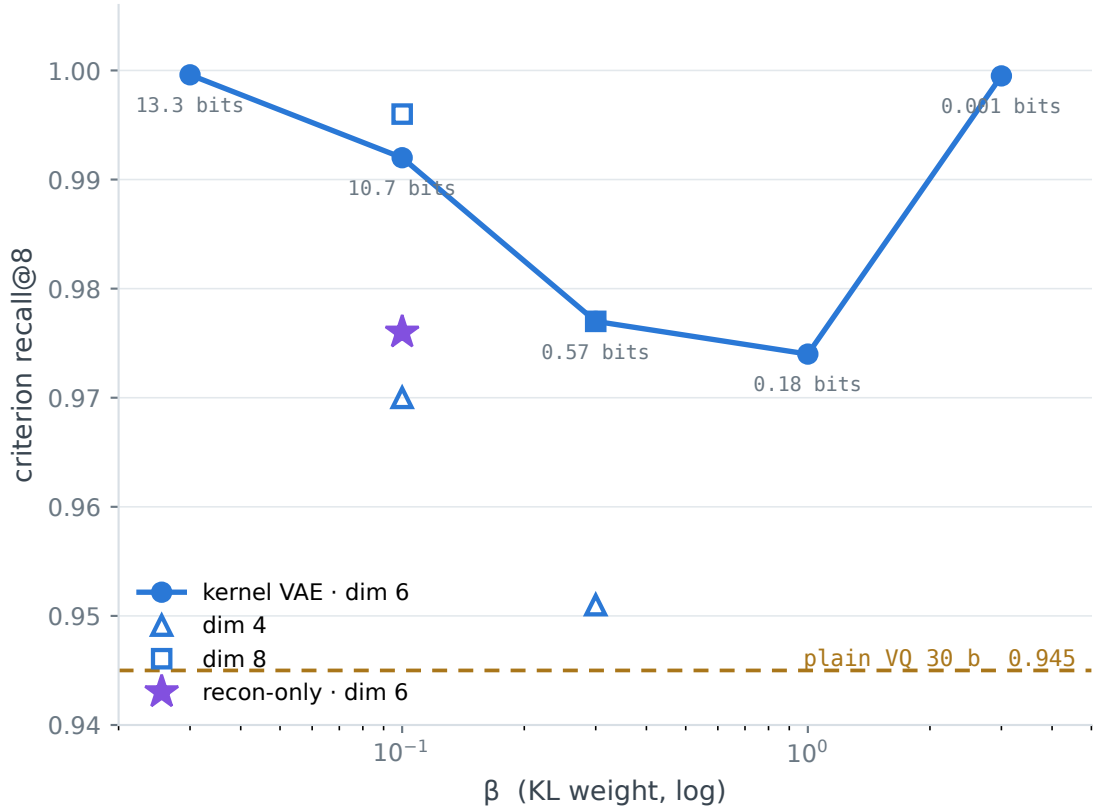


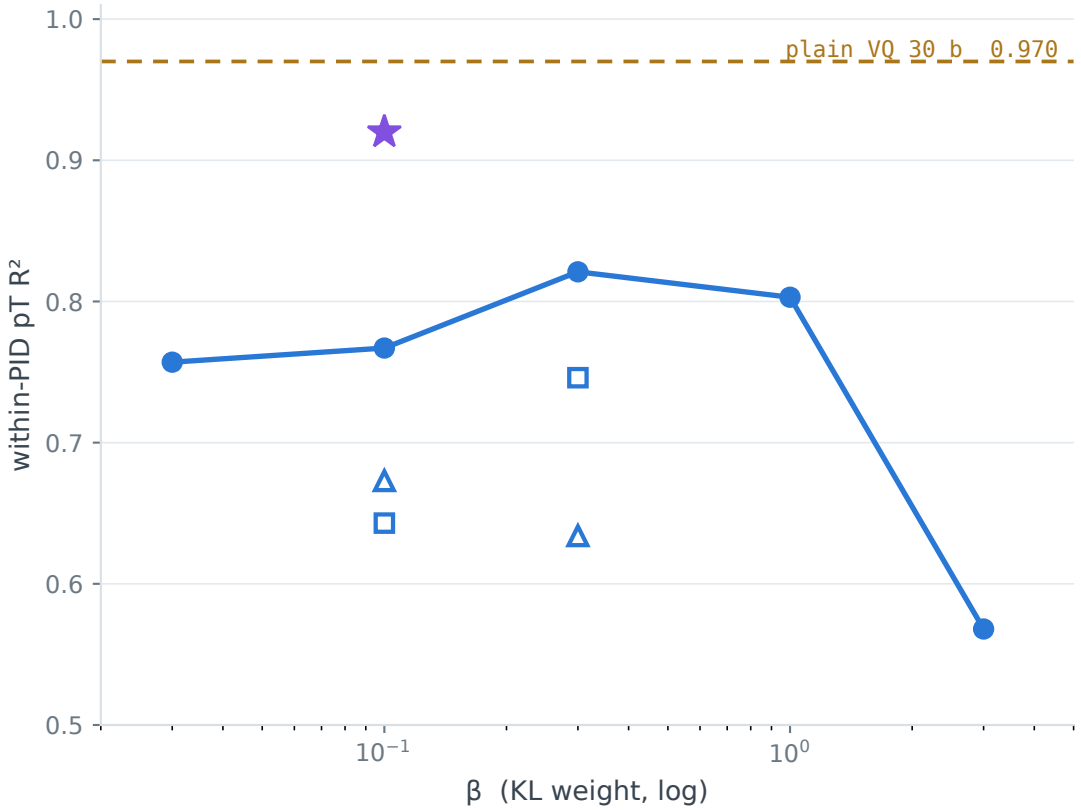
The kernel's fixed point drives criterion partners toward coincidence in the latent; any quantizer maps coincidence to identical codes. Kernel arms occupy ~2 % of the grid vs ~28 % for reconstruction.

Retrieval cannot rank true partners among thousands of identical-code impostors, so recall collapses. The mechanism is quantizer-independent — one cause explains the RVQ and FSQ negatives.

Implication: install the criterion on a continuous latent; a discrete tier should quantize the features (F5), not the criterion-shaped embedding.

F4 — the continuous token compresses; β trades information, never the criterion





Recall holds ≥ 0.95 across the whole dial while R² peaks at ≈ 0.82 (β = 0.3) — the information side never reaches VQ. Dimensionality: 16 → 8 is free, dim 6 is the sweet spot, dim 4 the floor; the encoder itself halves without loss (width 64 ≈ 256).

F5 — untrained references are strong; learned discrete tokens barely pay rent

DISCRETE INTERFACE	BITS	RECALL@8	PT R^2	PID PROBE
physics grid, double (untrained)	**26.3**	**0.9655**	0.949	1.00
physics grid, single (untrained)	23.3	0.9553	0.948	1.00
plain VQ 3×1024 (trained)	30	0.945	**0.970**	1.00
plain VQ 3×512 (trained)	27	0.927±.006	0.971	1.00
FSQ recon-only (trained)	24	0.851	0.840	0.81
library → post-hoc RVQ	27	0.832	0.174	0.38

- The zero-parameter criterion-aligned grid (ln(1.1) log-pT × 0.1 η/φ bins ⊕ exact PID/charge, half-bin-offset double grid) exceeds every learned discrete token on recall at equal or fewer bits.

- The trained VQ's entire surviving advantage is +0.02 pT R^2 and +0.017 purity — and the grid is computable on any detector, making it a natural cross-experiment release tier.

evidence: campaign 6b vs 2b/4/5/6c discrete arms · the 192-b null plays the same role for continuous tokens (F2)

F6 — measurement caveats that changed recorded conclusions

KL-rate is not a release rate for deterministic-μ tokens

At β = 3, measured KL ≈ 0.001 bits yet recall = 0.9995: μ is deterministic and cosine retrieval is scale-invariant, so structure survives at 'zero' ELBO rate. Continuous rates are now quoted as fp16 dim×16; the earlier '~1.4-bit' result is relabelled.

Objective conclusions must be rate-controlled

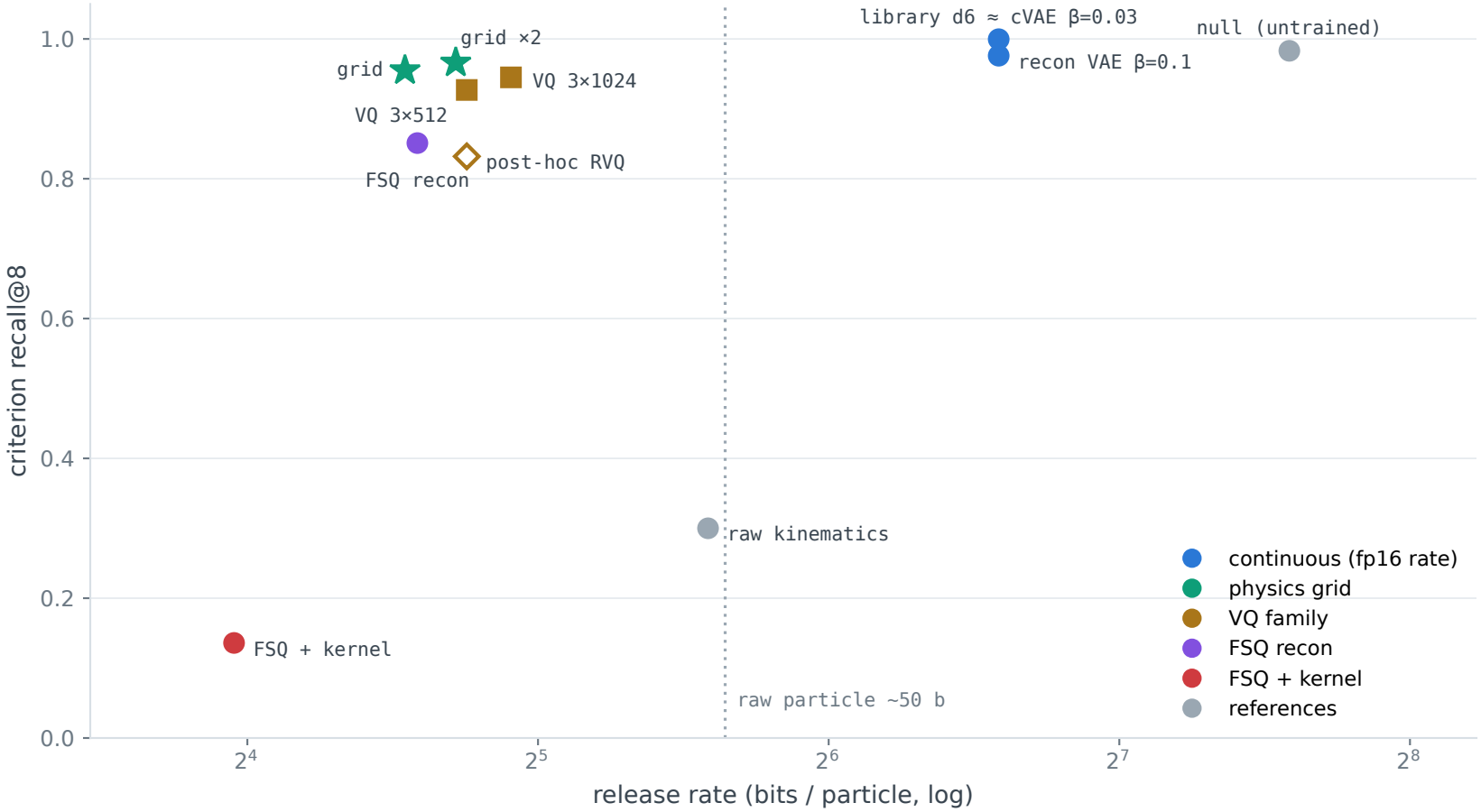
'Reconstruction buys species, not the criterion' (recall 0.383) was a β = 1 artifact: at β = 0.1 the recon-only VAE reaches 0.976 ± 0.004 with R^2 0.920 — still below the null, but the original causal story was wrong.

Seed discipline

Measured seed std is ±0.006 recall; single-seed deltas of similar size (e.g. context helping the 30-bit VQ by +0.022) are flagged, and headline claims are promoted only at 3 seeds.

evidence: campaign 6a systematics · benchmarks/README.md §Campaign 6, findings 2-3

All release candidates on one axis (honest rates)



Continuous tokens at their fp16 release size (F6 convention); the 13.3 KL-bits of the β=0.03 arm is a diagnostic. Full map with every campaign-1–6 arm: benchmarks/figures/library_rate.png.

What works — ranked release candidates

#	ARTIFACT	ROLE	KEY NUMBERS
1	**contrastive VAE β=.03 / dim-6 library**	the criterion carrier — compact	recall 0.998–0.9996 @ 96 fp16 b; R^2 0.76
2	**hybrid kernel + recon, w=3**	best single-token compromise	recall 0.9955, R^2 0.908 @ 256 b
3	**physics grid, double (untrained)**	the discrete tier — any detector	recall 0.966, R^2 0.949 @ 26.3 b
4	**plain VQ 3×1024**	max-fidelity discrete, if needed	recall 0.945, R^2 0.970 @ 30 b

- Caveats per rank: (1) species structure is metric, not linear — nonlinear consumers only; (2) misses the corner by 0.004 recall and is the largest artifact; (3) within-bin detail is lost and its R^2 is by construction; (4) buys only +0.02 R^2 over the grid at more bits, with the quantizer-0 collapse still unexplained.

- Recommended release: (1) as the primary continuous token, (3) as the discrete companion tier — the pair covers both consumer types at 96 + 26 bits.

What to test next — ranked by decision value

#	EXPERIMENT	WHAT IT DECIDES	EFFORT
1	**masked-continuous FM probe**	which export do FM consumers prefer? the mission's own metric	runs today
2	**REPA frozen-teacher distillation**	is the unreached corner an optimization artifact? (frozen target)	small
3	**factorized release: symbols ⊕ latent**	exact PID/charge channel: linear decodability by construction	small
4	**entropy codec for μ tokens**	turns fp16-vs-KL into an operational sub-raw rate claim (F6)	moderate
5	**d0/dz + PUPPI restoration; anomaly probe**	closes the audit gaps: release completeness, mandated probe	small
6	**VQ hygiene (k-means init, resets, SimVQ)**	was plain VQ underestimated by q0 collapse? (plain arm only)	small

– Deprioritized by F3's quantizer-independent mechanism: Gumbel/dVAE bottlenecks and flow tokens (raw-parity rate). Data-blocked: cross-experiment invariance — the north star; needs a second-detector sample (a synthetic-response prototype can only falsify).

4 · RANKINGS

Closed directions — do not revisit without new mechanism

DIRECTION	REASON
criterion-shaped discrete tokens	any quantizer, any rate, both placements — code degeneracy (F3)
jet-context encoders for the released token	breaks the library property; destroys low-rate tokens (F1)
hard criterion windows	geometry kinks at window edges; purity 0.66 vs 0.91 (F1)
single large codebooks (8192×1)	collapses to 1.7 % utilization (batch-size confound noted)
post-hoc quantization of criterion tokens	0.832 vs 0.927 trained at 27 b — trained defends geometry
KL-bits as a release-rate claim	decouples from deterministic-μ information content (F6)

Each closure is evidence-bounded, not dogma: the condition that would reopen it is a genuinely different mechanism (as FSQ was for the codebook explanation — tested, and the closure broadened).

Conclusions

- The release decision reduces to a stated, quantified trade — criterion geometry vs kinematic information vs bits — and we recommend shipping two tiers: the dim-6 contrastive/library token (96 fp16 b, recall ≈ 0.999) and the untrained double physics grid (26.3 b, R^2 0.949, computable on any detector).

- Installing the physics criterion requires a smooth continuous latent. Quantizing a criterion-shaped embedding fails for a quantizer-independent reason (code degeneracy) — this direction is closed.

- Untrained references did the heaviest epistemic lifting: the 192-b null and the 26-b grid bound what learning is actually buying, and every future scheme must beat them first.

- Rate claims must be operational (fp16 today, an entropy codec next); ELBO rates are diagnostics.

- The single most decision-relevant missing number is consumer-side: the masked-continuous FM probe over the release menu is queued as the next experiment.

Outlook and material

- North star (data-blocked): cross-experiment invariance — same-physics particles from CMS and a second detector sharing one library. The context-free encoder and the grid tier are both constructed for it; a second-detector sample is the missing input.
- All numbers in this talk are reproducible from the repository: benchmarks/README.md (campaign record, per-campaign tables), benchmarks/run_campaign*.sh (idempotent sweeps), eval_library.py (metric suite), docs/research/ (method of record; campaign-6 design, pre-registration and adversarial audit).
- Per-campaign detail (setups, full tables, seed studies) is available as backup material on request.

References

VQ-VAE	van den Oord, Vinyals, Kavukcuoglu — Neural Discrete Representation Learning, arXiv:1711.00937
FSQ	Mentzer et al. — Finite Scalar Quantization: VQ-VAE Made Simple, arXiv:2309.15505
BSQ / LFQ	Zhao et al., arXiv:2406.07548 · Yu et al. (MAGVIT-v2), arXiv:2310.05737
VQ gradient fixes	rotation trick, arXiv:2410.06424 · SimVQ, arXiv:2411.02038
REPA / RAE	Yu et al., arXiv:2410.06940 · arXiv:2510.11690
continuous AR tokens	MAR, arXiv:2406.11838 · GIVT, arXiv:2312.02116 · JetFormer, arXiv:2411.19722
HEP tokenization	OmniJet-α, arXiv:2403.05618 · MPM, arXiv:2401.13537 · tokenization-free MPM, arXiv:2409.12589
SSL objectives	VICReg, arXiv:2105.04906 · JetCLR, arXiv:2108.04253 · HEP-JEPA, arXiv:2502.03933 · J-JEPA, arXiv:2412.05333
per-particle representations	EFN/PFN, Komiske, Metodiev, Thaler — arXiv:1810.05165
data	CMS Aspen Open Jets, RunG 2016 open data release
this work	benchmarks/README.md (campaign record) · docs/research/ (method of record, C6 design & audit)